

Keyfob Transmitter

2-to-6 button remote control for gates and alarms — transmitter mode
— 433.92 MHz with EETX433A

Application guide for the most common use of the EETX433A: a handheld keyfob remote of 2 to 6 buttons for gates, barriers and alarm panels, with a unique code and ultra-low power.

FREQUENCY

433.92 MHz

RF POWER

13 dBm

BUTTONS

2 to 6

BATTERY

CR2032

VERSION

v1.0

BASED ON IC

EETX433A

DEVELOPED BY

Engeletron Engenharia

DATE

2026-07-04

• Introduction

This application note describes the most common use of the **EETX433A**: a **keyfob transmitter** — the handheld 2-to-6 button remote used to operate **gates, barriers and alarm panels**. The IC works in transmitter mode (Mode 0): each button sends a unique RF code that the gate or alarm receiver recognizes and acts upon.

The EETX433A integrates a 433.92 MHz PLL oscillator, OOK/ASK modulator, power amplifier and the encoding and power-management logic in a single SOP-8 package. So the remote needs only the IC, a few passives, the antenna loop, the buttons and a coin cell (CR2032). The **800 nA** standby current gives years of battery life.

Being 433.92 MHz RF (not infrared), it works **without line of sight**, through walls and doors and at a distance. The IC electrical details (pinout, absolute ratings, operating conditions, frame format and code mapping) are given in the EETX433A datasheet and referenced throughout this document.

• Operating Principle (Mode 0)

Mode 0 (transmitter) is selected automatically: pins CFG1, CFG2 and CFG3 are left **open** at power-up (no tie to VDD/GND). In this mode those pins also act as direct key inputs (KEY1–KEY3) and the SENIN/KEYx pin participates in a matrix for keys 4 to 6.

Pressing a button momentarily ties the corresponding key input to VDD (logic 1): this wakes the IC, which reads which key was pressed, encodes the corresponding 4-bit value and transmits. Up to **3 buttons** connect directly (one per CFG pin); buttons **4, 5 and 6** are added through the KEYx pin matrix (see the schematics).

Transmission continues **while the button is held**, with a minimum of **864 ms** (16 packets) and a maximum of **5,400 ms** (100 packets). If the user releases before 864 ms, the IC still completes the minimum. The LED blinks during transmission. With no button pressed, the IC is in ultra-low-power standby (800 nA) — the coin cell lasts for years.

• **Button Configurations**

The EETX433A covers 2 to 6 buttons on the same pinout. Up to 3 buttons connect directly (one per CFG1/CFG2/CFG3 pin); buttons 4, 5 and 6 are added by a simple matrix using the KEYx pin. The four reference schematics below (2, 3, 4 and 6 buttons) share the same RF core — 433.92 MHz loop antenna, matching capacitors, indicator LED and the CR2032 battery — changing only the number and wiring of the buttons.

Direct buttons (1–3)

Each button ties its CFG pin (CFG1/CFG2/CFG3) to VDD when pressed. It is the simplest wiring, enough for most 1-to-3-function gate keyfobs.

Matrix buttons (4–6)

The extra buttons share the KEYx pin (SENIN/KEYx) in a matrix with the CFG pins, reaching 6 keys with no extra pins. Each key still transmits a unique 4-bit value.

• Button Configurations

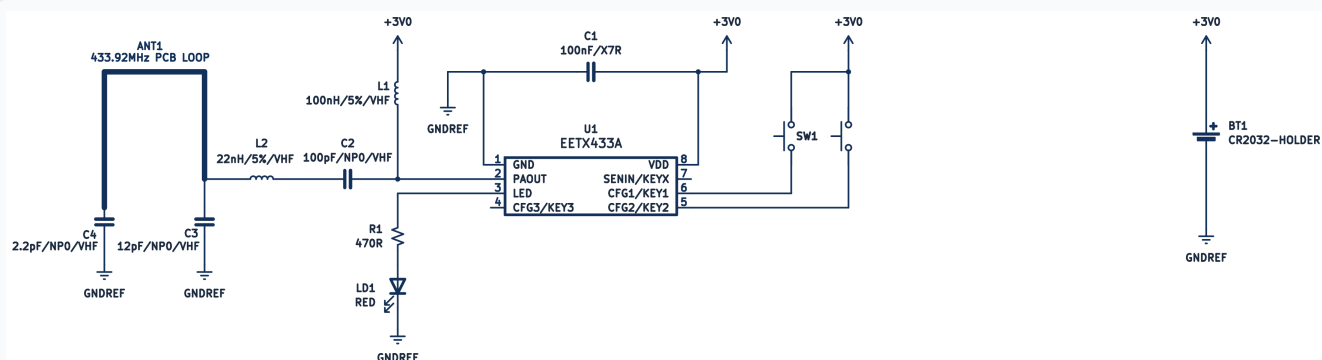


Figure 1: 2-button remote — KEY1 and KEY2 direct to the EETX433A

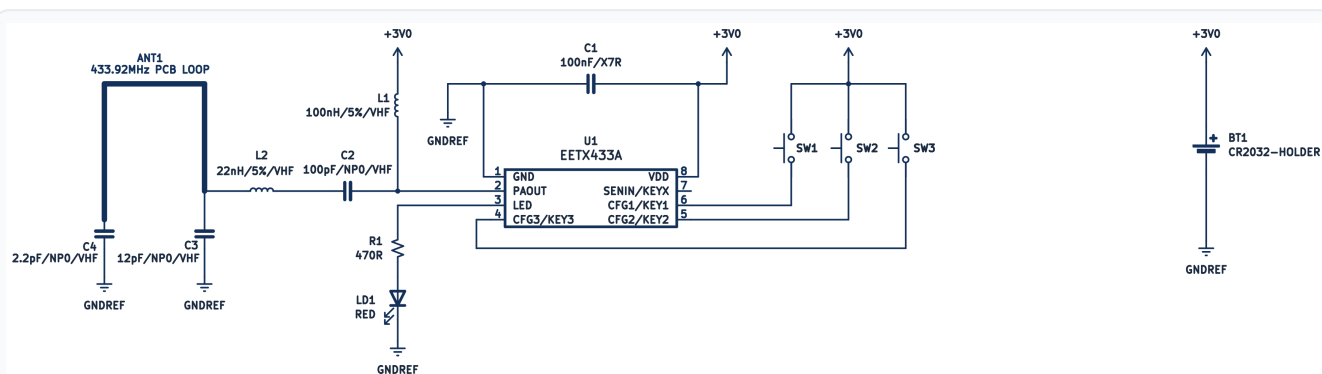


Figure 2: 3-button remote — KEY1, KEY2 and KEY3 direct

• Button Configurations

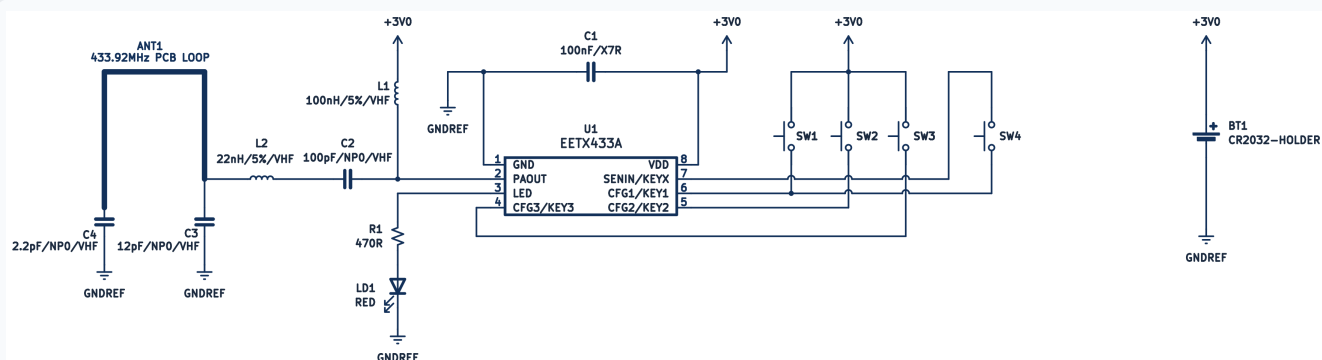


Figure 3: 4-button remote — 3 direct + 1 through the KEYx matrix

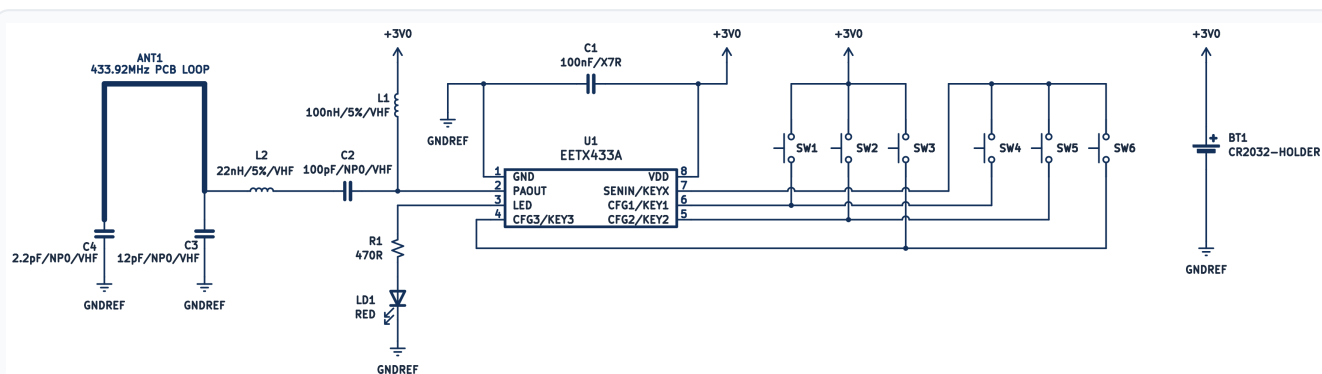


Figure 4: 6-button remote — 3 direct + 3 through the KEYx matrix

• Key Mapping

Each button transmits a fixed 4-bit value (D3D2D1D0) in the data field. The receiver identifies the pressed button from this value:

Key	D3	D2	D1	D0	Decimal
KEY1	0	0	0	1	1
KEY2	0	0	1	0	2
KEY3	0	0	1	1	3
KEY4	0	1	0	0	4
KEY5	0	1	0	1	5
KEY6	0	1	1	0	6

The 4-bit value identifies the key; the unique 20-bit address (factory-programmed) identifies the remote. Thus several remotes with identical buttons are distinguished by address at the receiver.

• Operating Sequence

Transmission starts when a button is pressed and lasts while the button is held, within the firmware time limits. The LED indicates each stage.

Phase	Condition	Behavior	LED
Idle	No button pressed	Ultra-low-power standby (800 nA) — battery lasts years	Off
Trigger	Press a button	Wake-up ~50 ms + PLL ~0.2 ms; key read	On
Transmission	Button held	Sends the key code; minimum 864 ms (16 packets)	Blinks on transmit
Maximum duration	Button held for a long time	Transmission ends at 5,400 ms (100 packets)	Blinks to the end
Early release	Release before 864 ms	The IC still completes the minimum time (864 ms)	—

GUARANTEED MINIMUM TIME

Even a quick tap guarantees the minimum transmission of 864 ms (16 packets), enough for the receiver to safely decode the command. Holding the button extends transmission up to a maximum of 5,400 ms.

- **Transmitted RF Signal**

Transmission follows the EETX433A packet format: pulse-width OOK encoding at 433.92 MHz, basic time unit $\lambda = 500 \mu\text{s}$ (each bit takes $3\lambda = 1.5 \text{ ms}$). Each packet carries pilot, start, 20 bits of programmable address (factory-unique), 4 bits of data (the key) and 4 bits of fixed anti-code, totaling $108\lambda \approx 54 \text{ ms}$, repeated 16 to 100 times depending on how long the button is held.

The 20-bit address uniquely identifies the remote and the 4-bit field identifies the key (see Key Mapping). The fixed anti-code improves decoding robustness at the receiver. See the EETX433A datasheet for the full frame structure and code mapping.

• Technical Characteristics

Values for the complete remote (EETX433A IC + buttons + battery), in transmitter mode. All RF and timing parameters follow the EETX433A datasheet.

Parameter	Value
Operating frequency	433.92 MHz
Channels	1
Modulation	00K / ASK
RF power (typ.)	13 dBm
Buttons supported	2 to 6
Supply voltage	2.0–3.6 V (typ. 3.0 V)
Typical battery	1× CR2032 (3 V)
Standby current	800 nA
Transmit current	12 mA
Minimum transmission	864 ms (16 packets)
Maximum transmission	5,400 ms (100 packets)
Wake-up time	50 ms typ / 100 ms max
Code per key	4 bits (1–6)
Address	20 bits, factory-unique
IC package	SOP-8

Battery life is dominated by standby (800 nA) and usage frequency; with normal use a CR2032 lasts years. Transmit current at 13 dBm per the EETX433A datasheet.

• Application Guidelines

- **Battery:** use a CR2032 cell (3 V) with a holder; the ultra-low standby (800 nA) gives years of life. There is no need to switch the remote off.
- **Buttons:** for 2–3 keys, wire each button directly to its CFG pin. For 4–6 keys, follow the KEYx matrix schematic (Figures 3 and 4). Use good-quality tactile buttons for durability.
- **Receiver learning:** each remote has a unique address; enroll it into the gate/alarm receiver using the learn procedure (receiver "learn" button + one press of the remote). Several remotes can be enrolled into the same receiver.
- **Antenna:** use the printed 433.92 MHz loop antenna per the reference schematic; keep the matching capacitors (C2/C3/C4) near the output pin and avoid ground plane and metal over the antenna.
- **Range:** position the antenna away from the user's hand and from batteries/metal; a vertical loop orientation usually gives the best range.
- **Layout:** keep the RF traces short and the ground plane continuous under the digital section, following the layout guidelines in the EETX433A datasheet.

• Soldering and Reflow

Excess temperature can damage the EETX433A, causing thermal deformation of the package and a shift of the 433.92 MHz frequency. The reflow limit is **≤ 210 °C for ≤ 10 s** — the component's own maximum rating, more restrictive than the standard J-STD-020 reflow classifications (220–235 °C for Sn-Pb).

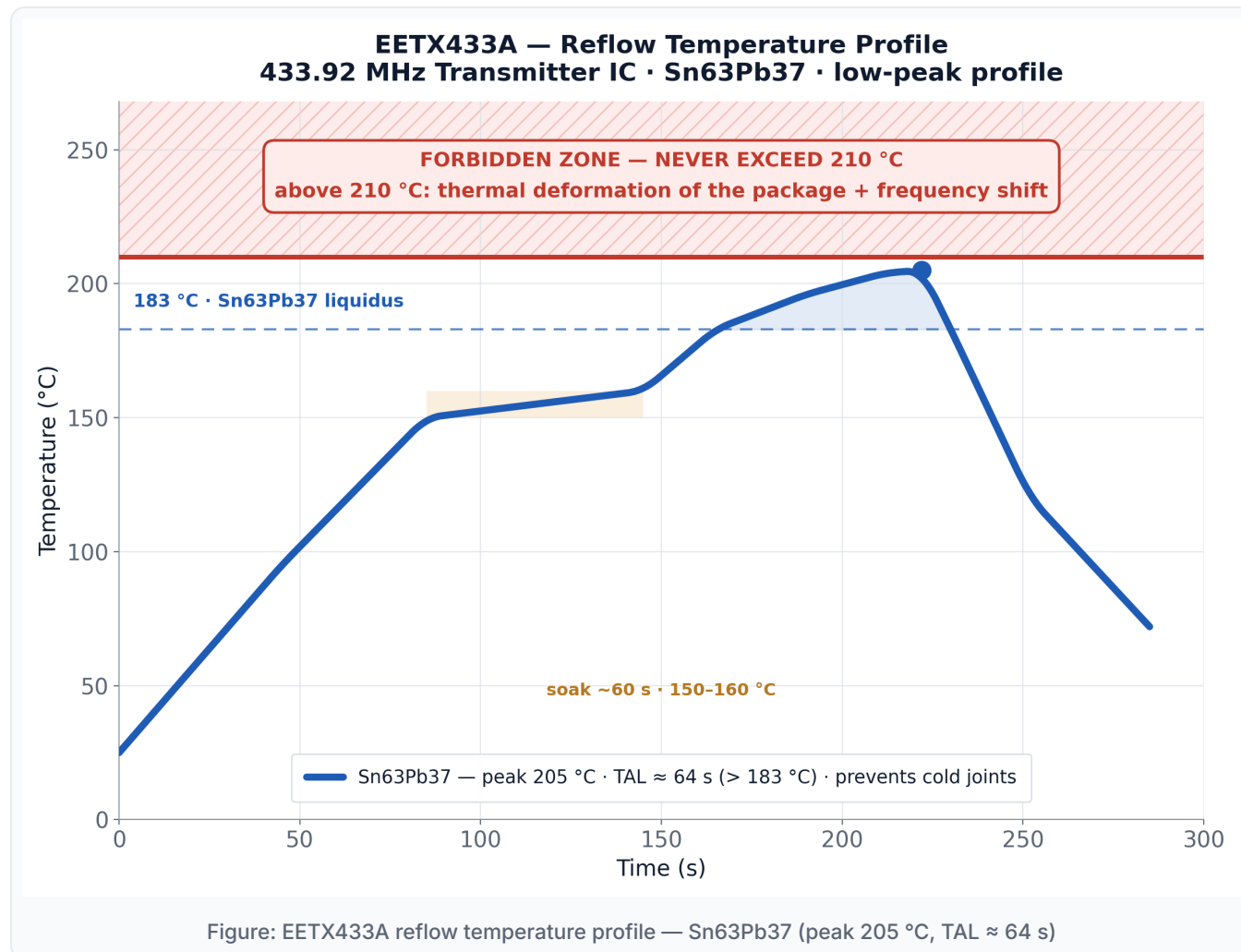
Alloy: **Sn63Pb37** (eutectic, 183 °C liquidus — the market standard), whose reflow peaks typically at **205–220 °C**; for the EETX433A the peak stays at the low end: board peak **205–210 °C** (target ~205 °C, never above 210 °C), compensated by a **TAL of 60–75 s**. Because the peak is modest, the time above liquidus is extended — and it is this combination that forms the intermetallic without exceeding the component limit.

To avoid **cold joints**: keep the soak of ~60 s between 150–160 °C (activates the flux and evens the board) and the **TAL of ~64 s above 183 °C** (ensures full melt and good wetting). Ramp-up ≤ 2.5 °C/s and cooling ≤ 4 °C/s. A high, short peak is not needed: a low peak plus an adequate TAL produces the same joint, safely for the IC.

NARROW WINDOW — MEASUREMENT IS MANDATORY

The Sn63Pb37 window here is only ~5 °C wide: **below 205 °C** there is a risk of cold joints; **above 210 °C** the IC is damaged (SOP-8 deformation and frequency drift). The profile **must be measured with a thermocouple on the board**, at least once per line. NEVER exceed 210 °C.

• Soldering and Reflow



Phase	Time (s)	Temperature (°C)
Ramp-up	0–85	25 → 150
Preheat (soak)	85–145	150 → 160
Ramp to peak	145–212	160 → 204
Peak (reflow)	212–228	205 (max. 210)
TAL — time above liquidus	≈ 64	> 183
Cooling	228–285	205 → 72

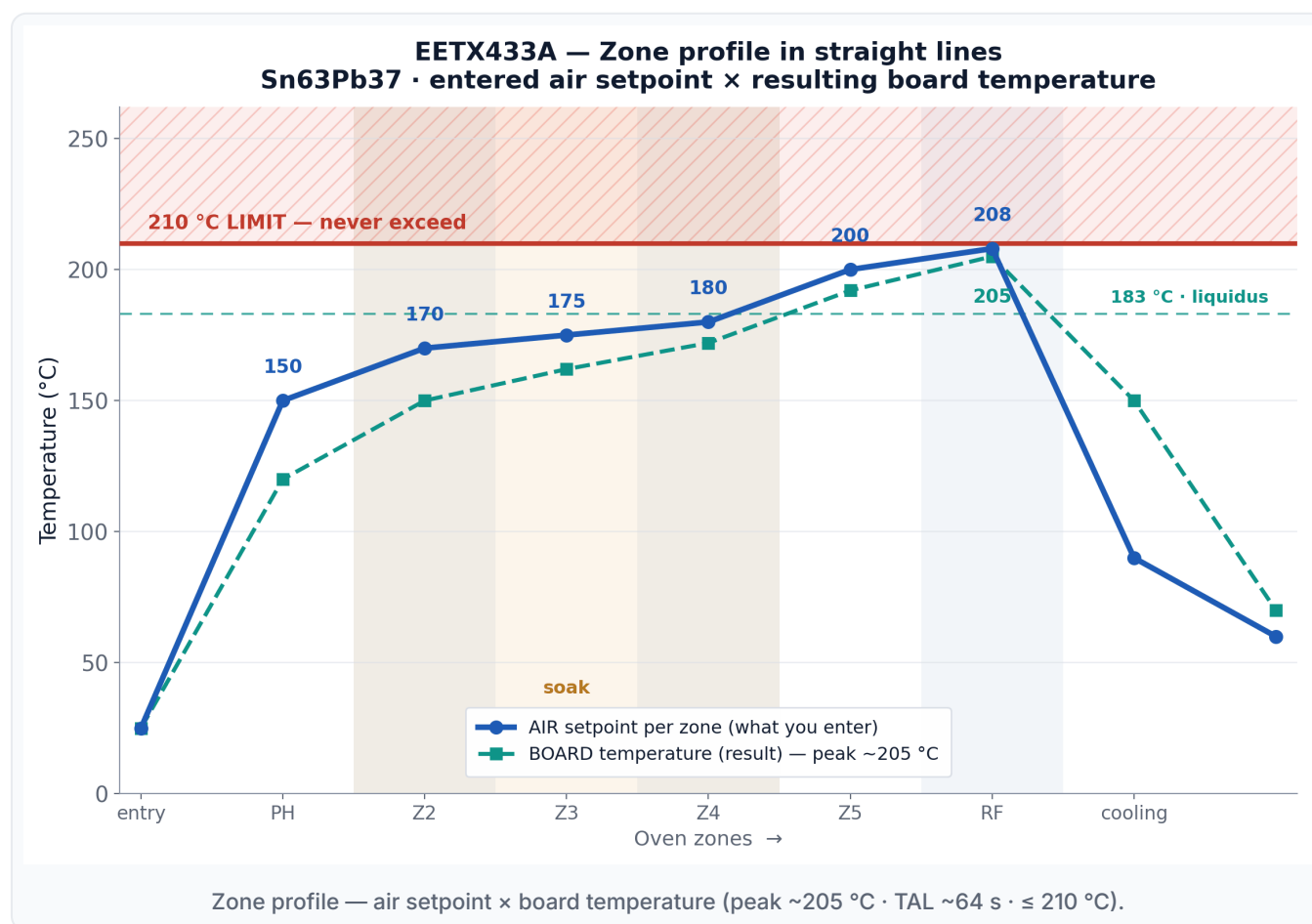
Sn63Pb37 (eutectic, 183 °C liquidus) — market standard. Low-peak profile: peak 205–210 °C (never above 210 °C), compensated by a TAL of 60–75 s. A ~60 s soak at 150–160 °C and a ~64 s TAL prevent cold joints; ramp-up ~1–2.5 °C/s · cooling ≤ 4 °C/s · max. 2 reflow cycles. Hand soldering (samples): tip ≤ 300 °C, contact ≤ 5 s.

• Practical oven setup — zone profile

The chart below shows the reference setup for a tunnel oven: the **air setpoint per zone** (solid line — 150 · 170 · 175 · 180 · 200 · 208 °C) and the resulting **board temperature** (dashed). Air setpoints are a starting point — tune by profiling with a thermocouple on the board. What matters is the board result: **peak ~205 °C and TAL ~64 s**, never above 210 °C.

NARROW WINDOW — MEASUREMENT IS MANDATORY

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To learn more about the EETX433A and obtain the full datasheet, visit: engeletron.com.br

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